

HS123

24/7/25

Need not be moral

THEORY OF RATIONAL CHOICE

Decision maker chooses that action which is the best to her according to her preferences from all available options.

$$A = \{a_1, a_2, \dots, a_n\}$$

Set of actions available to decision maker

$a_i > a_j \implies a_i$ is strictly preferred to a_j

$a_i \sim a_j \implies$ Indifference

From consistency of choices if

$$\underbrace{a > b, b > c}_{\Downarrow} a > c$$

Transitivity

Rational choice does not mean person has to be selfish, can take into account how his actions affect others.

PAYOFF FUNCTION (Utility function)

$$u: A \longrightarrow \mathbb{R}$$

If $a > b$ then $u(a) > u(b)$; $a, b \in A$

Payoff function gives ordinal preference information.

$$a > b > c$$

Means both of these could be valid payoff functions

①

$$u(a) = 100$$

$$u(b) = 2$$

$$u(c) = 0$$

②

$$v(a) = 1$$

$$v(b) = 0.99$$

$$v(c) = 0.98$$

$$u_i(a_1, a_2, \dots, a_{i-1}, a_i, a_{i+1}, \dots, a_n)$$

n individuals

Payoff function also depends on the actions taken by other individuals too.

Shortcut

$$u_i(a_i, a_{-i})$$

$$a_j \in A_j \quad \forall j \in 1, 2, \dots, n$$

$$\vec{a} = (a_1, a_2, \dots, a_n)^T$$

action profile

vector

STRATEGIC GAMES (Simultaneous move games)

Elements of the Game:

① Player set: Set of decision makers.

$1, 2, 3, \dots, n$; $n \geq 2$

→ Need at least 2 players

② Set of actions : A_i

Eg: For the game of chess, which is sequential and not strategic on the basis of the given definition, there are 20 possible first moves.

③ Preferences : of each player over the action profile have to be defined.

$$u_i : A_1 \times A_2 \times A_3 \dots \times A_n \longrightarrow \mathbb{R}$$

(a) PRISONER'S DILEMMA

Actions		Prison Sentences	
Prisoner 1	Prisoner 2	Pr 1	Pr 2
Confess (C)	Confess (C)	3	3
NC	NC	1	1
C	NC	0	5
NC	C	5	0

1. Players : Pr 1, Pr 2

2. Actions $A_1 = A_2 = \{C, NC\}$

Payoff function

Player 1

$$u_1(C, NC) > u_1(NC, NC) > u_1(C, C) > u_1(NC, C)$$

3 2 1 0

Player 2

$$u_2(NC, C) > u_2(NC, NC) > u_2(C, C) > u_2(C, NC)$$

Payoff Matrix / Table

		Pr 2	
		C	NC
Pr 1	C	1, 1	3, 0
	NC	0, 3	2, 2

$M_1(C, NC)$ ②
 $M_2(C, NC)$ ②
 $M_2(NC, C)$ ②
Row player payoff, Column player payoff

		NC	C
	NC	2, 2	0, 3
	C	3, 0	1, 1

(NC, NC) is **Pareto Superior** to (C, C)

Better for both parties

Improves the situation of at least one person without making the other person worse off

DUPPOLY is similar to prisoner's dilemma

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Free Ride: Optimal action when bus conductor is not paying

Eg:- attention is to not pay for the ticket.

Arms Race: Trying to develop superior technology under secrecy.

Eg:- ① Countries: 1, 2

② Actions: Build bombs, refrain

	BB	R
BB	1, 1	3, 0
R	0, 3	2, 2

③ Performance

Eg:- Prisoner's Dilemma between more than 2 players

TRAGEDY OF COMMONS

① n sheep farmers

② $A = \{ \text{Graze more, Graze less} \}$

③ $u_i(GM, GL) > u_i(GL, GL) > u_i(GM, GM) > u_i(GL, GM)$
vector of actions chosen by other villagers

action profile

	X	Y
X	2, 1	0, 5
Y	3, -2	1, -1

Q.

Prisoner's dilemma

Absolute value of payoff function does not matter, ordinal preference is imp. It is a prisoner's dilemma problem as the preference of both the players based on the payoff is the same

BATTLE OF SEXES (BOS)

Story: Husband and wife need to agree on a place to spend time together

① Players: H & W

② $A: \{ \text{Boxing match (B)}, \text{Opera (O)} \}$

③ $u_H(B, B) > u_H(O, O) > u_H(B, O) = u_H(O, B)$

$u_W(O, O) > u_W(B, B) > u_W(B, O) = u_W(O, B)$

	B	O
B	2, 1	0, 0
O	0, 0	1, 2

BOS Games

No specific preferences

1, 1	0, 0
0, 0	1, 1

Pure Coordination

Assurance

1, 1	0, 0
0, 0	2, 2

Coordination Games

Desirable outcome is to land on the diagonal

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2

PD:

		C	NC
I	C	1,1	3,0
	NC	0,3	2,2

BS:

		B	O
H	B	2,1	0,0
	O	0,0	1,2

Eg:- No possibility of cooperation

MATCHING PENNIES

① Players: 1, 2

② Actions: H, T

③ Preferences $u_1(H, H) = u_1(T, T) > u_1(H, T) = u_1(T, H)$

1
1
-1
-1

$u_2(H, T) = u_2(T, H) > u_2(H, H) = u_2(T, T)$

Payoff matrix

	H	T
H	1, -1	-1, 1
T	-1, 1	1, -1



Not possible to cooperate

Zero sum games / Strictly competitive games

STAG HUNT

① Players : 1, 2

② Actions : $\{St, H\}$ ← Pursuing

③ Preferences : $u_1(St, St) > u_1(H, H) = u_1(H, St) > u_1(St, H)$
2
1
1
0

Payoff matrix

	St	H
St	2, 2	0, 1
H	1, 0	1, 1

NASH EQUILIBRIUM (solution concept)

State of rest

Eg:-

		W	
		B	O
H	B	2, 1	0, 0
	O	0, 0	1, 2

Best action for husbands depends on what wife chose.

Based on the belief of one player of the other player's choice from previous experience

* Validation of the belief.

$$a^* = (a_1^*, a_2^*, \dots, a_n^*)$$

action profile a^* is Nash eq if $\forall i, a_i^*$ is at least as good as any other $a_i \in A_i$ given that a_i^* is being played by other players.

?? Missing

Ex 27.1

2

PD: 1

		C	NC
C	1, 1	3, 0	
NC	0, 3	2, 2	

⇒

$1+\alpha, 1+\alpha$	$3, 3\alpha$
$3\alpha, 3$	$2+2\alpha, 2+2\alpha$

$0 \leq \alpha \leq \frac{1}{2}$ Remain PD
 $\alpha > \frac{1}{2}$

$\alpha \rightarrow$ Deg of altruism

Let's say $\alpha = 1$

	C	NC
C	2, 2	3, 3
NC	3, 3	4, 4

$\rightarrow$ Nash equilibrium

Now $\alpha = \frac{1}{2}$

	C	NC
C	1.5, 1.5	3, 1.5
NC	1.5, 3	3, 3

} All are nash equilibria

Ex 27.2 (Seats in a bus)

BOS Game (Solutions using Nash eq.)

BOS

	B	W	O
H B	2,1	0,0	
O	0,0	1,2	

Nash eq

If there

Nash eq.

If there are multiple nash eq, the predictive power of Nash eq, diminishes

2

	X	Y
X	1,1	0,0
Y	0,0	1,1

Pure coordination

Nash eq.

2

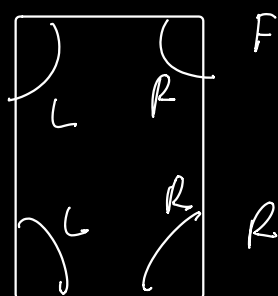
	X	Y
X	2,2	0,0
Y	0,0	1,1

Assurance

Nash eq.

Which type gets punished?

	FL	FR	RL	RR
FL	1,1	0,0	0,0	0,0
FR	0,0	1,1	0,0	0,0
RL	0,0	0,0	1,1	0,0
RR	0,0	0,0	0,0	1,1



Multiple eq

FOCAL POINT

Matchings Pennies

	H	T
H	1, -1	-1, 1
T	-1, 1	1, -1

No NE

Stag Hunt

	St	H
St	2, 2	0, 1
H	1, 0	1, 1

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	X	Y
X	(1,1)	(0,0)
Y	(0,0)	(2,2)

Although both are Nash eq
(1,1) is the state where the
game terminates too although it
is less desirable.

QWERTY PARADOX

Generalising Stag Hunt to n Players

$\frac{1}{n}$ th of stag > 1 hare

n hunters to hunt 1 stag

Nash Equilibria:

- ① $(st, st, \dots, st)$
- ② $(H, H, \dots, H)$

Prove that above 2 are the only Nash equilibria

Each player gets $\frac{1}{n}$ th of the stag. If a player i chooses H he only catches the hare and his payoff decreases.

Show that there exists no other NE exists

Argue that if at least one guy goes after the hare
(INCOMPLETE)

①
Modified Stag Hunt ($m < n$ hunters are required to hunt the stag)

NE stays the same

~~Proof~~

② Modified ($m < n$ hunters only required)
 $\frac{1}{k}$ th of a stag } Have
but
 $\frac{1}{k+1}$ th of a stag } Have

$$n > k > m$$

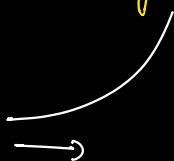
NE

① (H, H, ..., H)

② k stag hunters : $n C_k$

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Hawk - Dove Game (Game of Chicken)



Who chickened out first?

Players: Two fighters jits

Actions: Continue moving forward (aggressive), Reverse (passive)

Payoffs: $u_1(A, P) > u_1(P, P) > u_1(P, A) > u_1(A, A)$

③
②
①
④

2

	A	P	
A	0, 0	3, 1	→ Nash equilibrium
P	1, 3	2, 2	

Eg: Story of 2 Navals, they clash at the door.

Players: 2 Navals

Actions: Early, Late

Preferences: $u_1(L, E) > u_1(E, E) > u_1(E, L) > u_1(L, L)$

2

	A	P	
A	0, 0	3, 1	→ Nash equilibrium
P	1, 3	2, 2	

PUBLIC GOODS

Issue: Mentality of people where they want a free ride i.e. not pay for building a public good but still use it.

Players: People who live in a locality (n)

Actions: Pay or not pay

At least ($k < n$) contributions should be there for the public good to be provided.

Pr - Public good is provided
 nPr - " " not " "

Preferences: $u_i(NP, Pr) > u_i(P, Pr) > u_i(NP, nPr) > u_i(P, nPr)$

Possible NE

- ~~①~~ $c > k$, c is contributions
- ② $c = k$
- ~~③~~ $c < k$ (Except for when $c = 0$)

Preference order

- ① (NP, Pr)
- ② (P, Pr)
- ③ (NP, nPr)
- ④ (P, nPr)

STRICT NE

a^* is a strict NE if $\forall i$,

$$u_i(a^*) > u_i(a_i, a_{-i}^*) \text{ for}$$

$$\text{each } a_i \neq a_i^*$$

Exercise: Guessing the 2-3rds of average

Is there any integer k where (k, k, k) action profile
that is NE?

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Public Good

$$u_1(c_1, c_2) = v_1(c_1 + c_2) - c_1$$

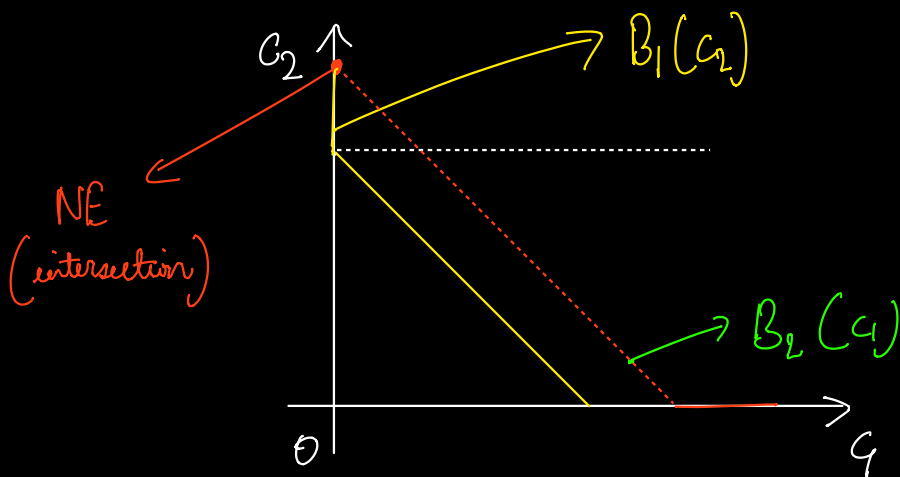
First order
condition

$$\frac{\partial u_1(c_1, c_2)}{\partial c_1} = 0 \implies v_1'(c_1 + c_2) = 1$$

$$\therefore c_1 + c_2 = v_1'^{-1}(1)$$

$$c_1 = v_1'^{-1}(1) - c_2 \\ = B_1(c_2)$$

$$\text{If } c_2 = 0, \text{ then } c_1 = v_1'^{-1}(1) \\ = B_1(0)$$



$$B_1(c_2) = v_1'^{-1}(1) - c_2 \quad \text{if } v_1'^{-1}(1) > c_2$$

$$= 0, \text{ otherwise}$$

Assumption

$$B_2(0) > B_1(0)$$

$$NE = (0, B_2(0))$$

Variant

$$\begin{aligned}
 u_1(c_1, c_2) &= (c_1 + c_2) + (w_1 - c_1) + (w_1 - c_1)(c_1 + c_2) \\
 &= w_1 + c_2 + (w_1 - c_1)(c_1 + c_2)
 \end{aligned}$$

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DOMINANCE OF ACTIONS

	No Jam	Jam
Bus	60	90
Bout	30	30

Bus strictly dominates Bout $\Rightarrow$ Bus is strictly dominated by bus
 Jam of commuters, assuming same cost

Eg :-

	L	C	R
T	1,1	1,2	1,1
B	1,0	1,2	1,3

} $\rightarrow$ C strictly dominates L, but not R.

2

PD :

	C	NC
C	1,1	3,0
NC	0,3	2,2

} C strictly dominates NC

W

		BOS	
		B	S
H	B	2,1	0,0
	S	0,0	1,2

$\implies$ No dominating action

profile in which

Q. Is the action a a strictly dominated action is played an NE?

Proof by contradiction: There exists a profitable deviation for at least one player to a strictly dominating action, therefore it is not an NE.

Definition of strict domination: For player i , a_i' strictly dominates a_i'' ($a_i', a_i'' \in A_i$) if $u_i(a_i', a_{-i}) > u_i(a_i'', a_{-i})$ for all possible profile a_{-i} .

WEAK DOMINATION

	No Jam	Jam
Bus	60	90
Bout	60	60

Bout weakly dominates bus $\implies$ Bus is weakly dominated by bus

Definition of weak domination: For player i , a_i' strictly dominates a_i'' ($a_i', a_i'' \in A_i$) if $u_i(a_i', a_{-i}) \geq u_i(a_i'', a_{-i})$ for all possible profile a_{-i} and there is at least one a_{-i} such that $u_i(a_i', a_{-i}) > u_i(a_i'', a_{-i})$.

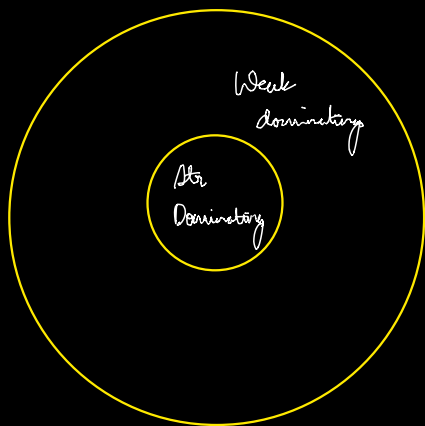
Q. Can a weakly dominated action be played in a strict NE? **No.**

Q. " " NE? **Yes**

Eg:-

	X	Y
X	0,0	0,0
Y	0,0	1,1

Y is a weakly dominating action which leads to an NE.



Q. Voting

mj-draft, Page 61

Candidates A and B

Odd no of voters $\rightarrow$ Players

Actions: A or B, no abstaining

Majority prefer A to B

Claim: Citizen voting for less preferred candidate is weakly dominated

① Case where the voter is indifferent, where the preferred candidate wins by 1 vote or more.

② Switching the vote makes a difference where the candidates have equal votes excluding the pivotal voter.

Finds all NE of this game.

Ans. All possibilities where margin of victory of A is odd and all the voters of A prefer that A would've won.

3 candidates in voting game

Q. Show that a citizen's only weakly dominated action is voting for the least preferred candidate. $A \succ B \succ C$

Must show that

there is at least one

scenario where voting A leads

to better outcome than voting C

similarly voting B leads to better outcome than voting C.

Voting A weakly dominates voting C but not voting for B.

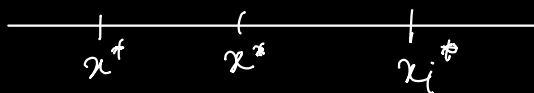
COLLECTIVE DECISION MAKING

Median of all votes is chosen

Setup

1. n citizens (odd)
2. deciding on a policy (x)

i's favourite policy x_i^* (Eg:- Allocating 1 lac Cr for Ed.)
i announces x_i ($x_1, x_2, x_3, \dots, x_n$)



$$x^* = \text{Median}(x_1, x_2, x_3, \dots, x_n)$$

$$u_i(a) > u_i(a') \text{ if}$$

The agent
of among the citizens
wants to be allocated

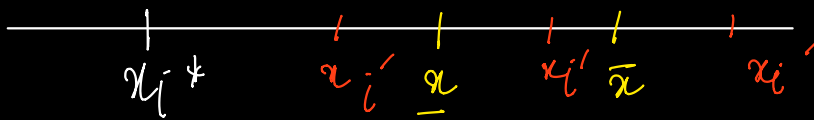
$$|x_i^* - \text{Med}(a)| < |x_i^* - \text{Med}(a')|$$

Q. x_i^* weakly dominates over all other x

Ans. Keeping i aside there are $n-1$ announcements. Let the 2 middle most observations be $\underline{x}$ and $\bar{x}$

Take $x_i > x_i^*$, $x_i = x_i'$

1, 1, 2, 3, 4, 5



Med if
 $x_i = x_i^*$

Med if
 $x_i = x_i'$

$$x_i^* < x_i' < \underline{x}$$

$\underline{x}$

$\underline{x}$

$$x_i^* \sim x_i'$$

$$x_i^* < \underline{x} < x_i'$$

$\underline{x}$

x_i

$$x_i^* > x_i'$$

$$x_i^* < \bar{x} < x_i'$$

$\underline{x}$

$\bar{x}$

$$x_i^* > x_i'$$

Symmetric case where x_i^* is $\bar{x}$ also behaves in a similar manner.

(Dominance solvable game)

OTHER APPROACHES TO GAME THEORY PROBLEMS

		C	NC ^x
PD	C	1,1	3,0
	^x NC	0,3	2,2

C strictly dominates NC for both players. $\therefore$ the choice (C,C) is the NE.

It is not necessarily true that choosing the dominating actions will result in an NE.

Doubtful

		L	C	R
T		1,1	0.5,1	0,0
M		1,0.5	2,2	0,3
B		0,0	0,1	1,0

No strict domination but there is still an NE.

Motivation of NE was that players' beliefs about the other players based on multiple rounds of the game is right.

NE needs a history of rounds of the game which is not always possible.

(source: Gibbons)

Iterated domination of or dominated action

	L	R
U	2, 2	1, 0
D	1, 0	0, 2

Assumption: Rationality is a common knowledge.

U is a dominating action; then the only remaining action profile is (U, L)

3rd solution concept (Minimax method) → Direct & Indirect

Only for zero-sum games

Each box contains the payoff of only player 1, the payoff of

Min player is -ve of that.

	L	M	R
T	2	5	13
H	6	5.6	10.5
L	6	4.5	1
B	10	3	-2

2 This is also the NE

5.6

Maximin = minimax = 5.6

1

-2

Solution: (H, M)

Max

10 5.6 -2

	L	C	R	Min
T	3	-2	2	-2
M	-1	0	4	-1
B	-4	-3	1	-4
Max	3	0	4	

4th solution concept (Rationalisability)



An action is said to be rationalisable if an action is the best action in response to an action

	C	NC	
PD			
C	1,1	3,0	
NC	0,3	2,2	

Solution is also an NE

For 1, is C rationalisable?

Yes, C is BR to both C & NC of 2.

Is NC rationalisable for 2?

No.

Is C of 2 rationalisable? Yes, it is BR to C of 1.

Source: Mas - Colell, Whinston & Green

Counterexamples where the solution is not an NE.

②

a b

c	1, 0	0, 1
d	0, 1	1, 0

①

a and b are justified for player 2 when player 1 plays d and c resp. This is true the other way around too.



so all actions

profiles are rationalisable, but none are NE.

SYMMETRIC GAMES & SYMMETRIC EQ.

SYMMETRIC GAMES

① 2 player game, $A_1 = A_2$

Same population group, set of actions is the same

$$② u_1(a_1, a_2) = u_2(a_2, a_1)$$

x y

x	c, c	b, a
y	a, b	d, d

①

SYMMETRIC EQ

	X	Y
X	0,0	2,2
Y	2,2	0,0

If eq takes place along the diagonal then it is justified, not in this case where it is along the off-diagonal.
meaning cannot be steady state

NE: (X,Y) and (Y,X) diagonal.

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Q.

	A	B	C
A	1,1	2,1	4,1
B	1,2	5,5	3,6
C	1,4	6,3	0,0

(a) Find all NE of the game?

(b) Which of the NE would be the steady state if it models pairwise interaction between members of a single population group? (A,A)

ILLUSTRATIONS OF NE IN STRATEGIC GAMES

(a) OLIGOPOLY MARKETS

COURNOT MODEL
BERTRAND

① Players: n producers ($n \geq 2$)

② Actions: $q_i \geq 0$, quantity of output is the action of player i ($i = 1, 2, \dots, n$)

③ Preference: Maximise their profit π

Assumption that in this market the manufacturer determines quantity and not price.

$$\pi_i(\cdot) = \text{Total revenue} - \text{Total cost}$$

$$= p(\text{price}) \times q_i - c_i(q_i)$$

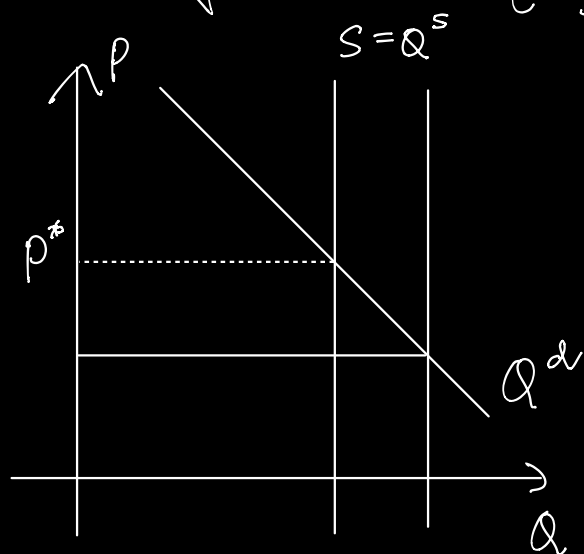
where $c_i(q_i)$ is the cost $\leftarrow$ cost of producing q_i units of the good

$$\text{Supply} = \sum q_i = Q^s$$

$$\text{Demand} = Q^d = f(p), \quad f'(p) < 0 \quad \leftarrow \text{demand curve}$$

$\hookrightarrow$ as price rises, less is bought

$\Rightarrow$ Inverse demand curve : $p = f^{-1}(Q^d)$



Market Eq. :- $Q^d = Q^s$

$$\therefore \sum_i q_i = f(p)$$

$$\therefore p = f^{-1}(Q^s)$$

$$\begin{aligned} \pi_i() &= \text{Total Revenue} - \text{Total cost} \\ &= p(\text{price}) \times q_i - c_i(q_i) \end{aligned}$$

$$= p q_i - c_i(q_i) = f^{-1}(\sum q_i) q_i - c_i q_i$$

$$c_i(q_i) = c q_i \quad ; \quad c > 0$$

Assumption

$$Q^d = \frac{\alpha - p}{d}$$

$$\alpha > c$$

$$d > 0$$

$\rightarrow$ Inverse demand curve : $p = \alpha - Q$, if $\alpha \geq Q$
 $= 0$, if $\alpha < Q$

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$$\pi_i(q_1, q_2)$$

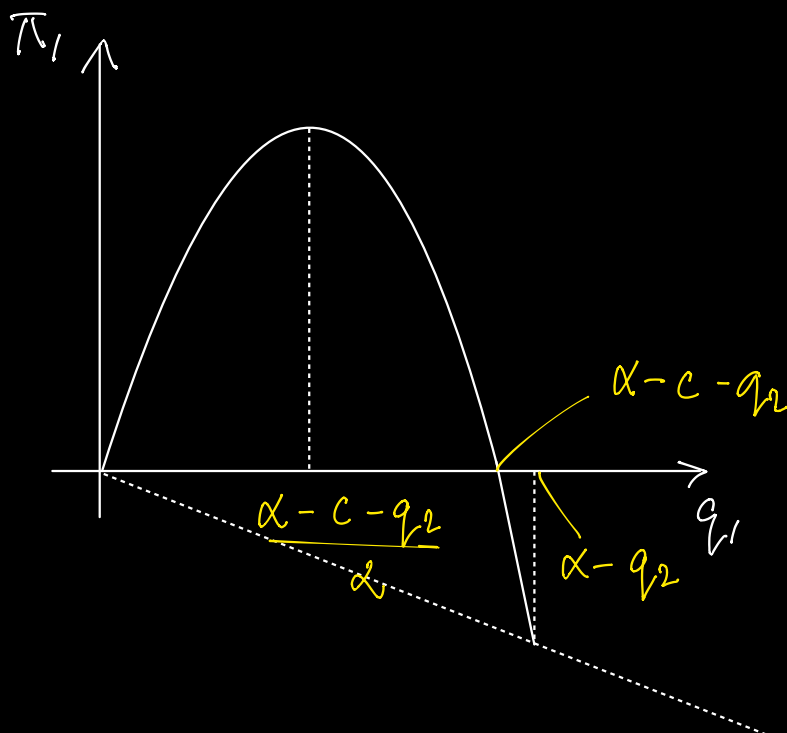
Cost function that
is preferred to be
maximised by
player i

$$\pi_i(q_1, q_2) = pq_1 - cq_1$$

$$= q_1(p - c) = q_1(\alpha - (q_1 + q_2) - c)$$

$$= q_1(\alpha - c - q_1 - q_2) \quad , \quad \alpha \geq q_1 + q_2$$

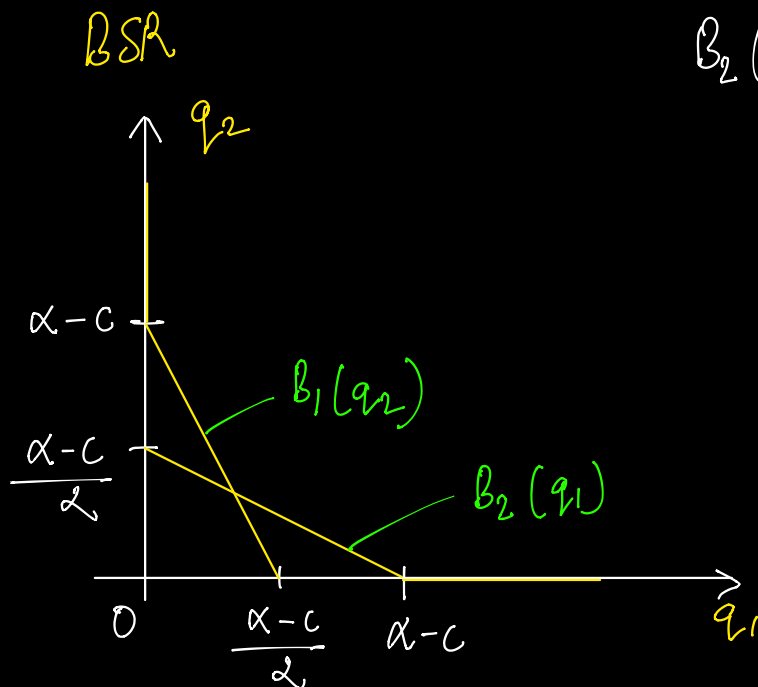
$$= -cq_1 \quad \text{if } \alpha < q_1 + q_2$$



Optimal q_1 , given q_2 ?

$$B_1(q_2) = \frac{\alpha - c - q_2}{2} \quad \text{if } q_2 \leq \alpha - c$$

$$= 0, \quad \text{if } q_2 > \alpha - c$$



$$B_2(q_1) = \frac{\alpha - c - q_1}{2}, \quad \text{if } q_1 \leq \alpha - c$$

$$= 0, \quad \text{if } q_1 > \alpha - c$$

$$q_1^* = \frac{\alpha - c - q_2^*}{2}$$

$$q_2^* = \frac{\alpha - c - q_1^*}{2}$$

$$(q_1^*, q_2^*) = \left(\frac{\alpha - c}{3}, \frac{\alpha - c}{3} \right)$$

$$p^* = \alpha - Q^* = \alpha - \frac{2}{3}(\alpha - c) = \frac{1}{3}(\alpha + 2c)$$

$$\pi_i(q_1, q_2) = p(q_1, q_2) q_i - c q_i$$

At NE

$$p^+(q_1, q_2) = \frac{\alpha + 2c}{3}$$

$$q_i^+ = \frac{\alpha - c}{3}$$

$$\therefore \pi_i^+ = \left(\frac{\alpha + 2c}{3} \right) \left(\frac{\alpha - c}{3} \right) - c \times \left(\frac{\alpha - c}{3} \right)$$

$$= \frac{\alpha - c}{3} \left(\frac{\alpha + 2c - 3c}{3} \right)$$

$$= \left(\frac{\alpha - c}{3} \right)^2$$

$$\pi_i^* = \frac{1}{9} - (\alpha - c)^2$$

Q. Why is it that when $\alpha \uparrow \Rightarrow q_i^* \uparrow, Q^* \uparrow, P^* \uparrow, \pi_i^* \uparrow$

Q. $c \downarrow \Rightarrow q_i^* \uparrow, Q^* \uparrow, P^* \downarrow, \pi_i^* \uparrow$

$\rightarrow$ unit cost of production

Due to better

means of manufacturing / technology

Q. $c_1 \neq c_2$ & what happens?

Above discussion is based on the assumption that producers want to maximise their profits, which is not always true. Eg:- Tio

Q. 2 firms

1st firm maximises market share

2nd firm " "

profits

Optimal s_1 is at $\pi_1(\cdot) = 0$

$$s_1(q_1, q_2) =$$

$$\frac{q_1}{q_1 + q_2}$$

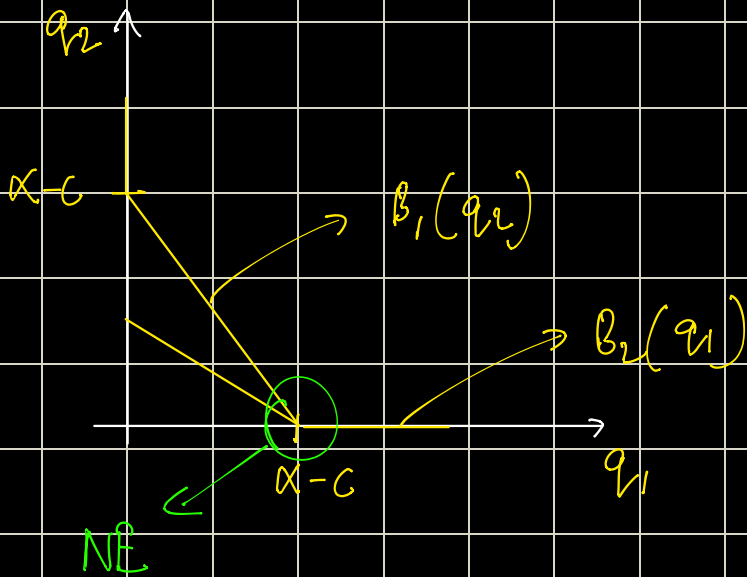
$$, \pi_1(\cdot) \geq 0$$

market share

$\rightarrow$ constraint that it does not make a loss

$$B_1(q_2) = \alpha - c - q_2, \quad \text{if } \alpha - c \geq q_2$$

$$= 0, \quad \text{if } \alpha - c \leq q_2$$



Explanation (Firm 2 is the market share maximizing firm)

$$\pi_2(q_1, q_2) = p(q_1, q_2) q_2 - c q_2 = q_2(p - c)$$

$$= q_2(\alpha - q_1 - q_2 - c) = q_2(\alpha - q_1 - q_2 - c)$$

COLLUSION

Maximising the joint profit by arriving at an agreement.

$$n=2$$

$$c_i(q_i) = cq_i, \quad P = \alpha - Q$$

Eg:- OPEC :- Organisation of Petroleum Exporting Countries

$$\pi = \text{Joint profit} = \pi_1 + \pi_2$$

$$= pq_1 - cq_1 + pq_2 - cq_2$$

$$= (p - c)(q_1 + q_2) = Q(\alpha - Q) - cQ$$

First order condition

$$\frac{d\pi}{dQ} = 0 \Rightarrow \alpha - Q - Q - c = 0$$
$$\Rightarrow \bar{Q} = \frac{\alpha - c}{2}$$

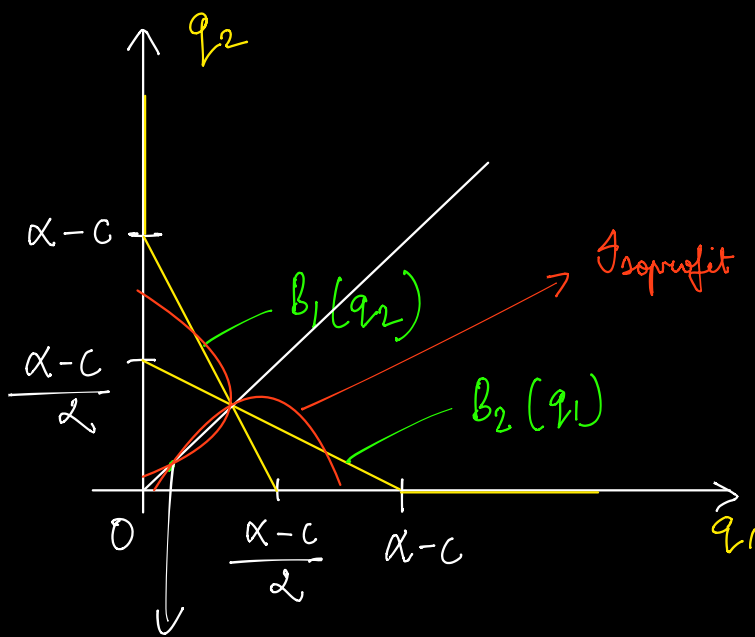
$$q_i^* = \frac{\alpha - c}{3} > \bar{q}_1 = \bar{q}_2 = \frac{\alpha - c}{4}$$

↑ price at the earlier NE

$$\bar{p} = \frac{\alpha + c}{2} > \frac{\alpha + 2c}{3} = p^*$$

$$\begin{aligned}\bar{\pi}_i &= \frac{\alpha - c}{4} \left(\alpha - c - \frac{\alpha - c}{2} \right) \\ &= \frac{1}{8} (\alpha - c)^2 > \frac{1}{9} (\alpha - c)^2 = \pi_i^*\end{aligned}$$

The above rate of production is not in the BSH of either player.



Collusion outcome
lies here, not at the BSH

Isoprofit curves: profit remains the same along these curves for the respective players.

At the lens of the curves (overlapping) the profit of the players are higher than NE

Above game is like PD, NE is (c, c) and collusion outcome is (NC, NC) .

Take $n \geq 2$ here

$$p = \alpha - Q$$

$$c_i(q_i) = cq_i$$

$$\alpha > c > 0$$

$$\pi_i(q_1, q_2, \dots, q_n) = pq_i - cq_i$$

$$= q_i \left(\alpha - \left(\sum_{i=1}^n q_i \right) - c \right)$$

$$(q_1^*, q_2^*, \dots, q_n^*) = \text{NE},$$

$$q_i^* = q_j^* \quad \forall i$$

From first order condition

$$0 = \alpha - c - 2q_1^* - q_2^* - \dots - q_n^*$$

$$0 = \alpha - c - q_1^* - 2q_2^* - \dots - q_n^*$$

$$0 = \alpha - c - q_1^* - q_2^* - \dots - 2q_n^*$$

$$q_1^* = \frac{\alpha - c}{n+1}, \quad q_n^* = \frac{n}{n+1} (\alpha - c)$$

$$p^* = \frac{\alpha + nc}{n+1}, \quad \text{as } n \rightarrow \infty, \quad p^* \rightarrow c$$

Perfect condition

Above game is similar to the tragedy of commons.

Ex 62.1

$$n \text{ firms, } u_i = \alpha_i (1 - (x_1 + x_2 + \dots + x_n)) \text{ if}$$
$$x_i = \text{resource used by } i \quad \sum x_j \leq 1$$